# 01 The life science tools framework

FRAMEWORK

A product framework spanning academic research, biotech, pharma, CRDMOs and manufacturing.

## Reagents

| | |
|---|---|
| 01 | Recombinant proteins / antibodies |
| 02 | Enzymes / PCR and NGS reagents |
| 03 | DNA / RNA / chemical building blocks |
| 04 | Cell lines / culture media |
| 05 | Assay reagents for R&D and QC |

Tests × dose per test × price

## Consumables

| | |
|---|---|
| 01 | Tips / microplates |
| 02 | Culture flasks / centrifuge tubes |
| 03 | Membranes / functional beads |
| 04 | Columns / chromatography resins |
| 05 | Gels / process components |

Tests or batches × intensity × price

## Instruments

| | |
|---|---|
| 01 | Sequencers / liquid handlers |
| 02 | LC / LC-MS / mass spectrometers |
| 03 | Flow cytometers |
| 04 | High-content imaging systems |
| 05 | Cell-culture workstations |

New installations + replacement + service

**Use one product taxonomy; retain the scope of each market estimate.**

Classify products separately from their use cases. Chromatography resins can be reused.

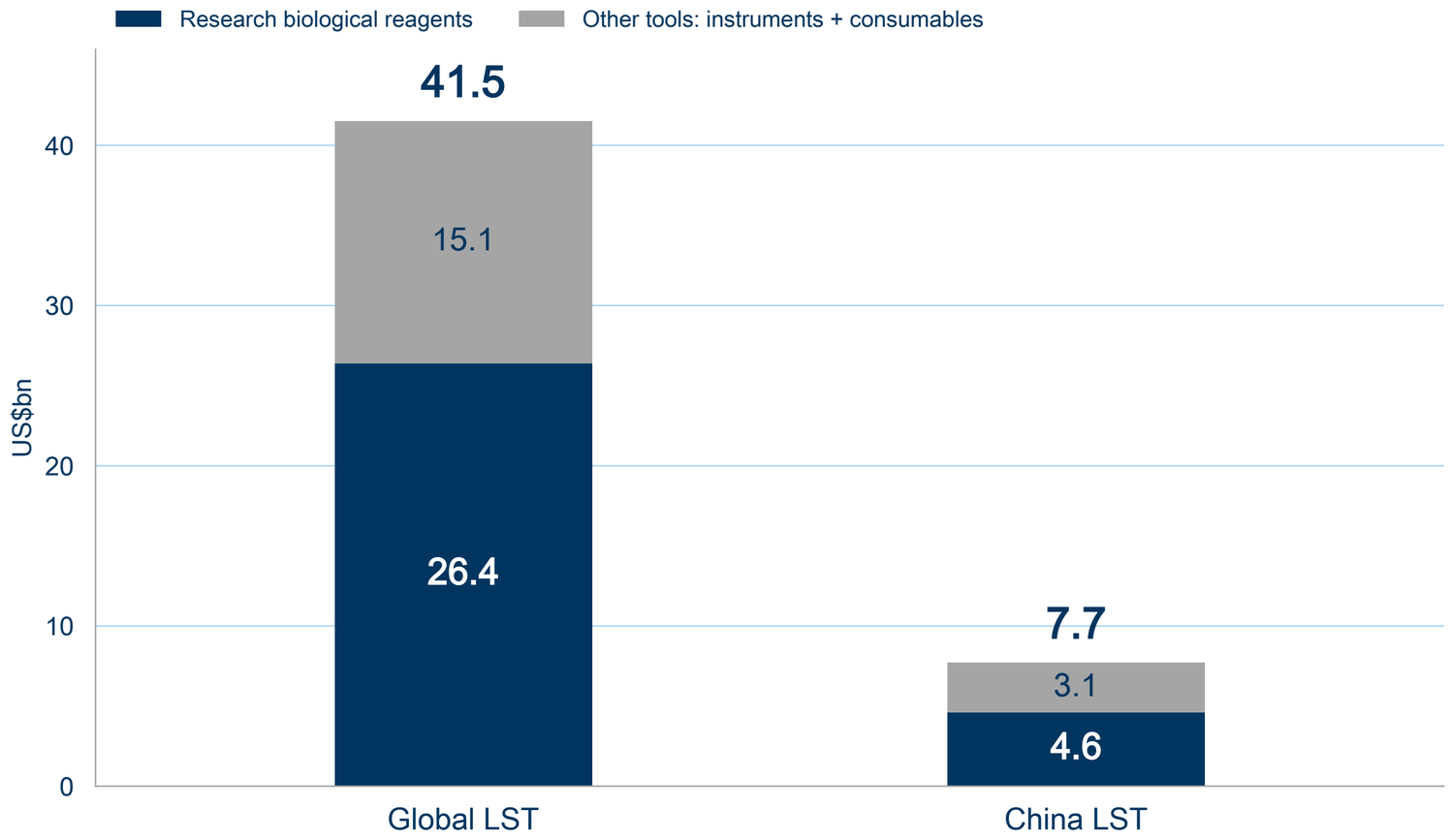
Source: supplied framework and prior product research. Existing TAMs do not automatically reconcile.

2026-10-12

# 02 2025 LST under one F&S definition

HISTORICAL

2025 F&S estimates commissioned for ACRO’s filing. A defined scope, not a broad-industry TAM.




China share within F&S scope

**18.6%**

7.7 / 41.5
Not vendor revenue exposure

Issuer-commissioned estimate.
Broader markets and vendor
China revenues use other scopes.

Other tools are the residual. Comparable instrument and consumable splits were not separately available.

Source: ACROBiosystems HKEX application, 30 Sep 2026, printed pp. 65 and 67; links in SOURCES.md.

# 03 Customer workflows across life science tools

USE MAP

A qualitative workflow map. Customer purchasing shares have not been established.

| CUSTOMER | REAGENTS | CONSUMABLES | INSTRUMENTS |
|---|---|---|---|
| **Academia / research hospitals** | Targets, enzymes, cell media | Plates, tips, cultureware | Sequencing, imaging, LC-MS |
| **Biotech / pharma R&D** | Proteins, assay panels, expression reagents | Sensors, columns, filtration | BLI / SPR, automation, flow cytometry |
| **CRO / CDMO** | Discovery and process reagents | Labware, resins, scale-specific inputs | Assay, process and QC platforms |
| **Biomanufacturing** | Production media, feeds and enzymes | Resins, filters, single-use assemblies | Bioreactors, chromatography, QC instruments |
| **IVD development** | Antigens, antibodies, tool enzymes | Assay plates, magnetic beads | Research and QC instruments |

**An instrument supplier’s total China sales are not its China life-science sales.**

Food, environmental and chemical-analysis uses excluded. Research instruments exclude clinical IVD analyzers.

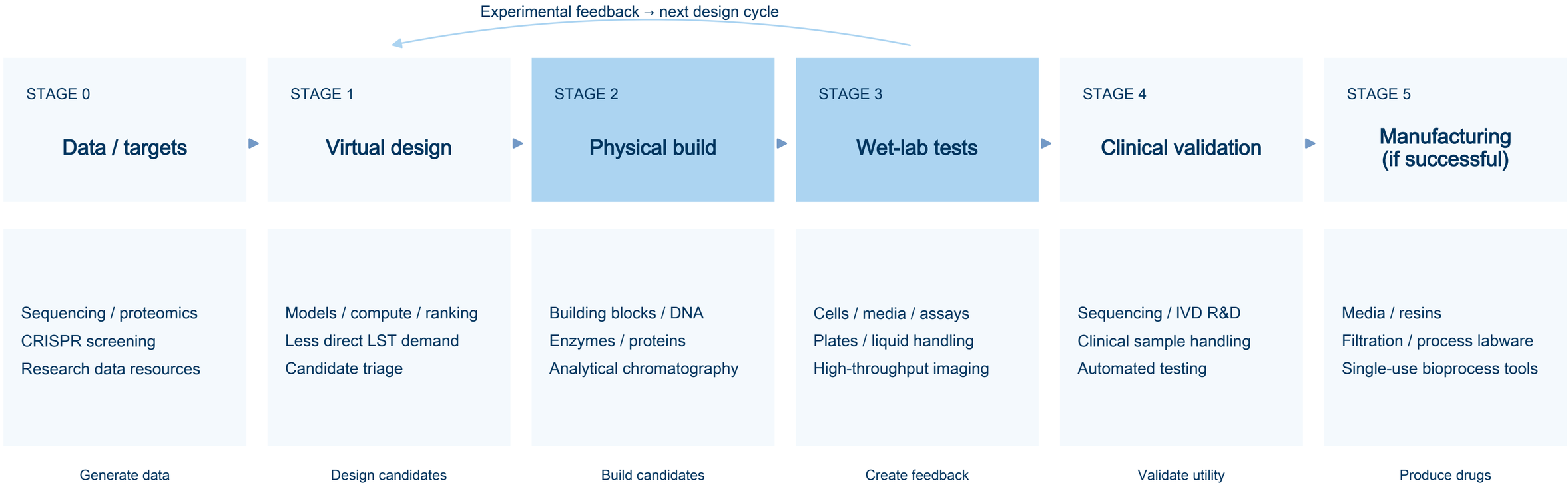
Source: representative supplier product uses; a role map, not an estimated customer revenue mix.

2026-10-13

# 04 AI discovery and its LST touchpoints

MECHANISM

AI candidates need physical construction and experimental feedback. Orders depend on funded projects.




**The most direct potential uplift comes from building and testing candidates.**

Stages 2–3 have more direct exposure. Clinical and manufacturing demand face attrition and time lags.

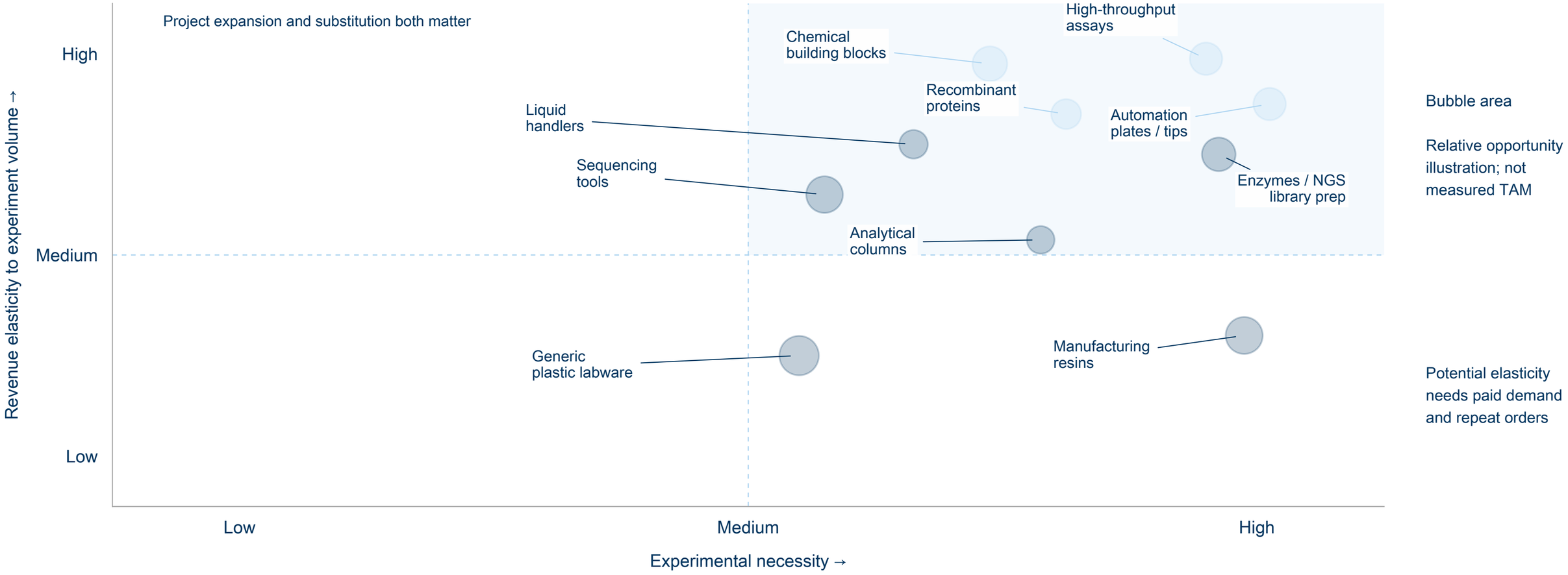
Source: supplied framework and product-use research. Arrows describe a mechanism, not realized orders.

2026-10-12

# 05 Which tools have more demand elasticity?

HYPOTHESIS

Positions and bubble areas are qualitative research judgments, not measured scores or additive TAMs.




Small-molecule synthesis can fall per project. Modality, assay depth, unit use and funded projects determine orders.

Source: supplied framework and analyst assumptions. Edit positions and relative areas in data.json.

2026-10-13

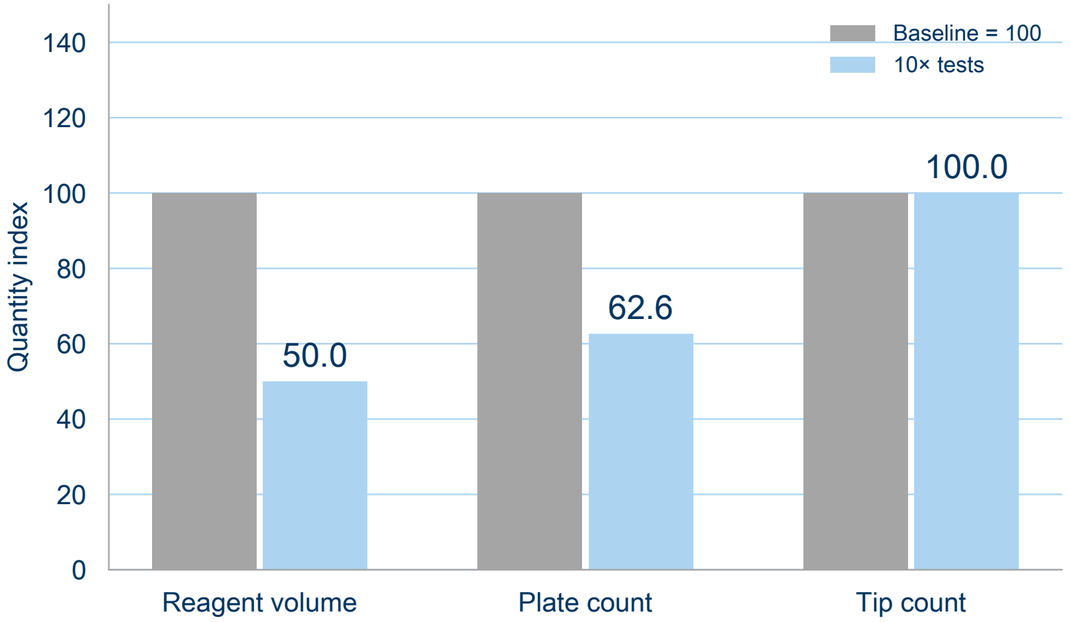
# 06 Quantities and prices in miniaturization

SENSITIVITY

10× tests: 10 L → 5 L reagent; 1,042 → 652 plates; 100k → 100k tips. Full usable plates.

## Quantities × net unit prices determine spending

### Physical inputs




### Fewer plates can cost more

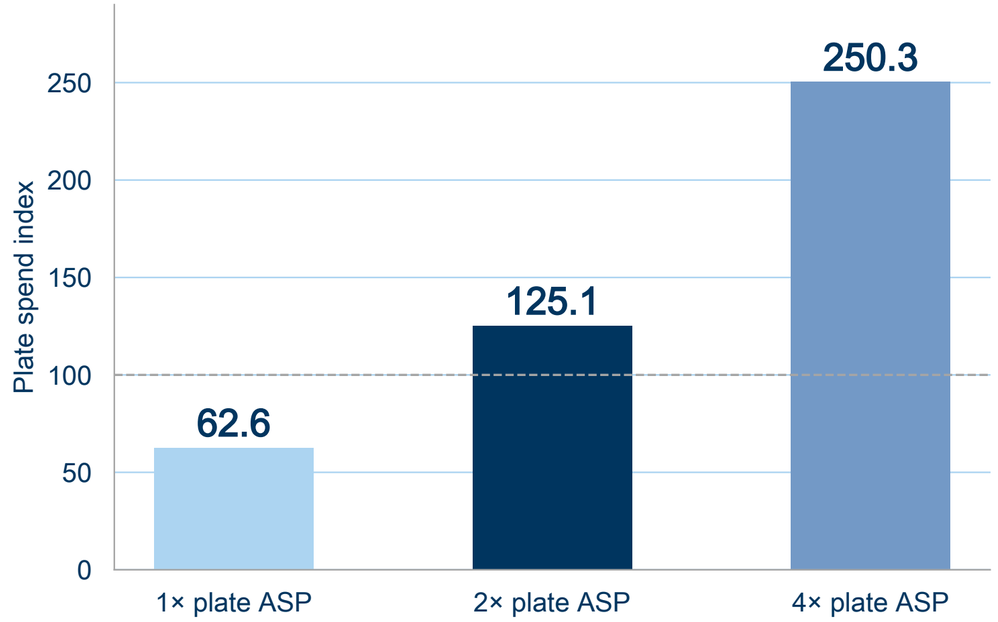



**Plate spending exceeds baseline if the plate ASP ratio is above 1.60×.**

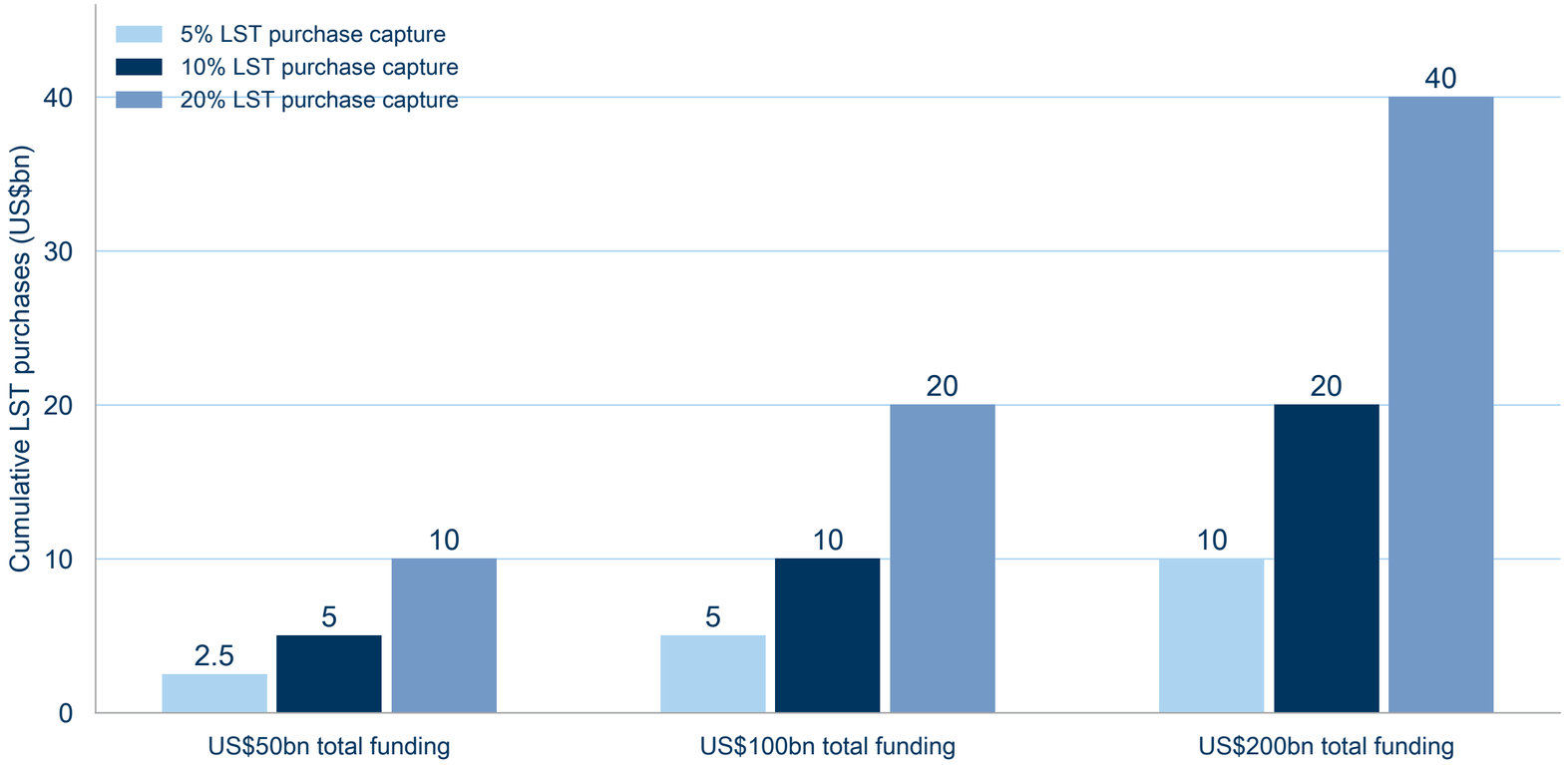
Price ratios are analyst inputs, not quoted ASPs. Dead volume, controls, usable wells and retesting can change quantities.

Source: transparent arithmetic from supplied test volumes and well counts. Units and sensitivities in SOURCES.md.

# 07 Funding-to-LST purchase sensitivity

SENSITIVITY

Hypothetical total funding pools. Only a fraction becomes purchases from LST suppliers.




## Funding → tool purchases

R&D and production budgets exclude people, trials, software and unspent cash.

Capture includes tool purchases only. Count CRO/CDMO inputs once, separate from service fees.

**Total capital raised is not an addressable tool market.**

5% / 10% / 20% capture rates are model inputs. Cumulative scenarios, no fixed horizon; not observed funding or annual TAM.

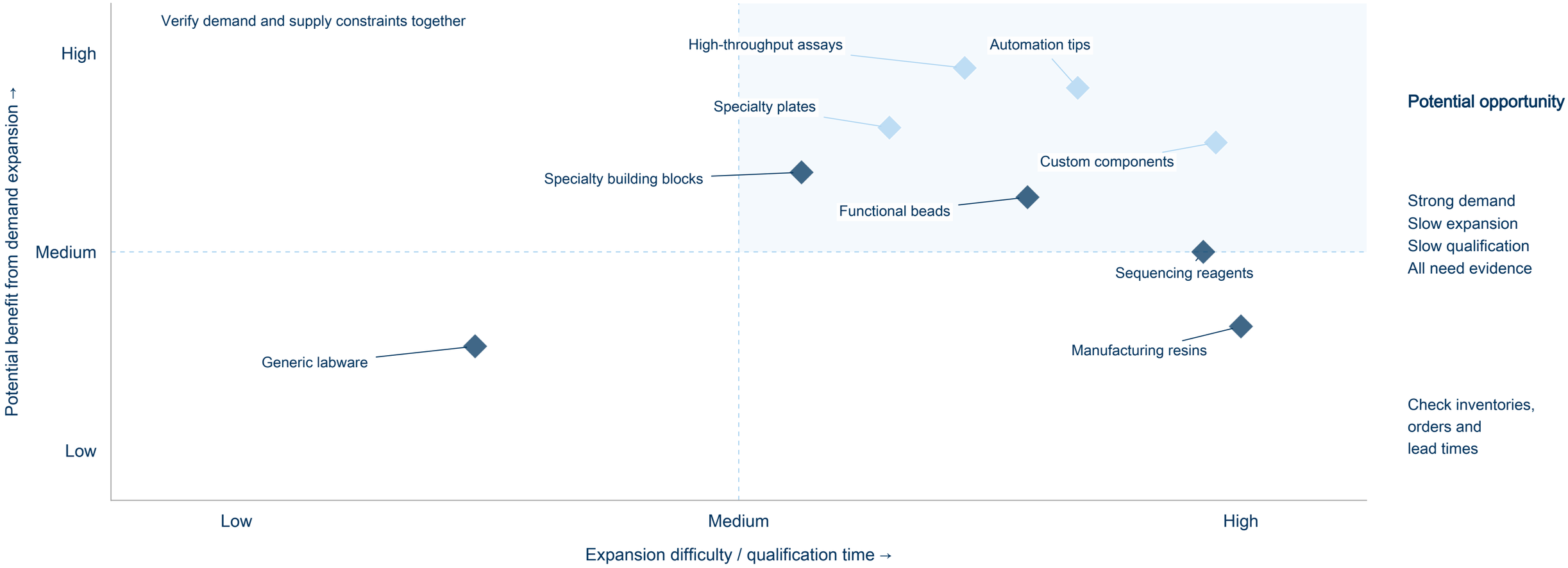
Source: analyst funding-to-purchase sensitivity. No empirical budget shares or equipment/reagent split is asserted.

2026-10-13

# 08 Potential supply constraints over 12–24 months

HYPOTHESIS

Relative expansion and qualification constraints over 12–24 months; current shortages are not established.




Track lead times, stockouts, qualification, yields and prices. Manufacturing resins have a longer demand lag.

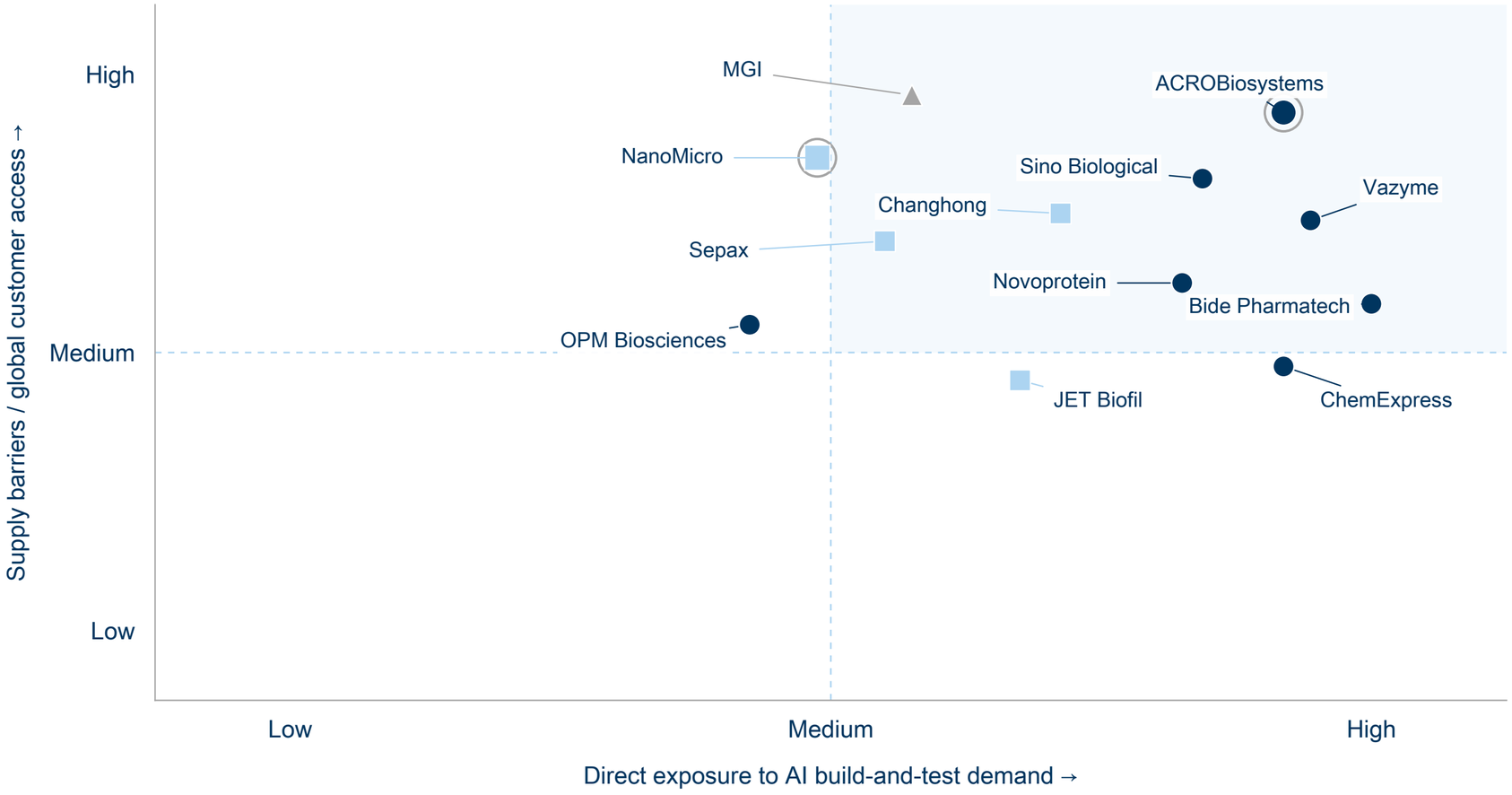
Source: supplied framework and analyst hypotheses. Technical barriers do not prove a capacity bottleneck.

2026-10-12

# 09 Mapping Chinese listed LST suppliers

HYPOTHESIS

Product exposure to AI build-and-test stages; positions do not measure AI revenue share or investment rank.




### Reagents

Proteins, enzymes, blocks, media

### Consumables

Labware, columns, resins

### Instruments / platforms

Sequencing and automation

### Two different demand pathways

ACRO: expression / functional assays
NanoMicro: purification + scale-up

Axes are qualitative judgments. Qualification, global customer access and paid orders still require evidence.

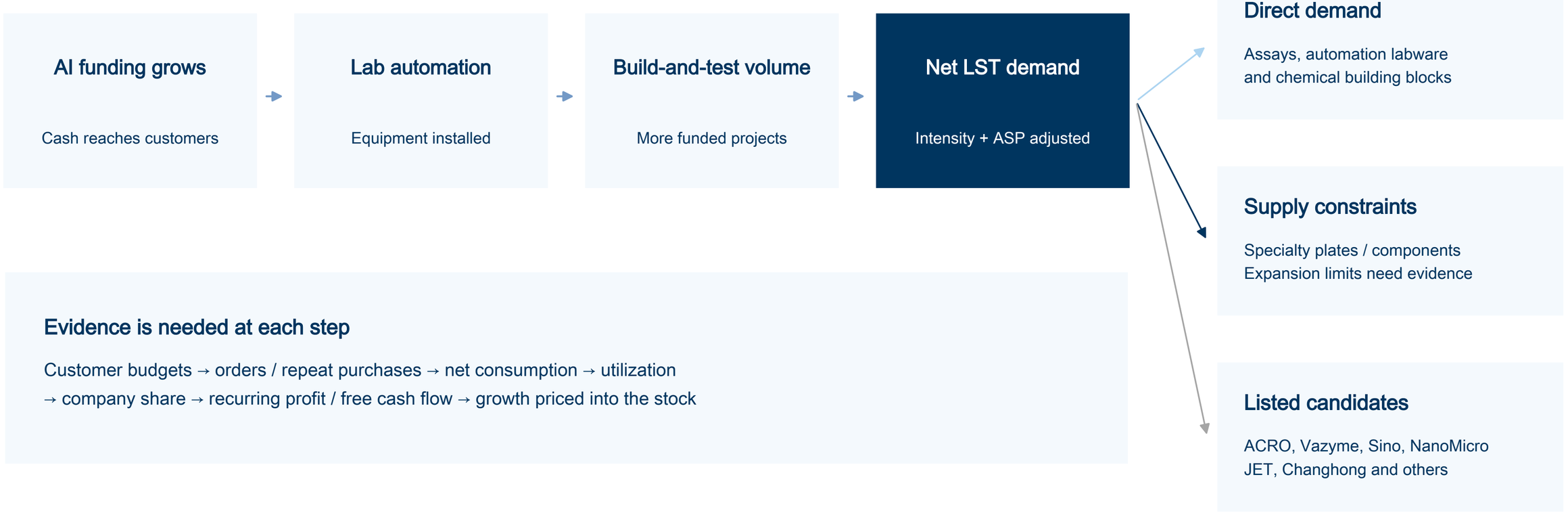
Sources: company filings and official product catalogs. Product rationale and assumptions are in data.json.

2026-10-12

# 10 From AI funding to LST shareholder returns

THESIS

Funding starts the chain. Projects, consumption, supply, market share and cash returns determine the outcome.




## Evidence is needed at each step

Customer budgets → orders / repeat purchases → net consumption → utilization
→ company share → recurring profit / free cash flow → growth priced into the stock

**Experimental relevance ∩ Evidenced supply constraints ∩ Global customer access**

Seek experimental relevance, evidenced supply constraints and global access; test the valuation hurdle.

Source: supplied framework and prior company research. Capital announcements are not LST orders.

2026-10-12

# AI discovery: workflows, evidence and sensitivities

Modalities use different units. Workload sensitivities and historical clinical benchmarks are shown separately.

| STAGE | BEFORE GEN-AI | AI MECHANISMS | WHAT CHANGES |
|---|---|---|---|
| Virtual search / design | Large virtual libraries | Quality + selection | Large libraries predate generative AI. The investment mechanism is better selection and learning. |
| Physical samples | Program-specific | Modality-dependent | Small molecules can need fewer syntheses per program. Protein libraries and individual profiling are different units. |
| **Assay workload** | **1× baseline** | **1× / 2× / 5× / 10×** | **Analyst workload sensitivities, not predicted fully profiled candidate counts. Include an efficiency/no-growth case.** |
| Clinical development | Phase I programs | Funded clinical programs | Clinical biology, patients, capital and time remain constraints; discovery gains do not establish clinical gains. |
| Phase I to approval | 7.9% / 13.8% (study estimates) | AI advantage unproven | BIO 2011–2020 and Wong 2000–2015 are different cohorts and methods, not an AI comparison. |

**Tool demand depends on funded programs, assay depth, net inputs and prices.**

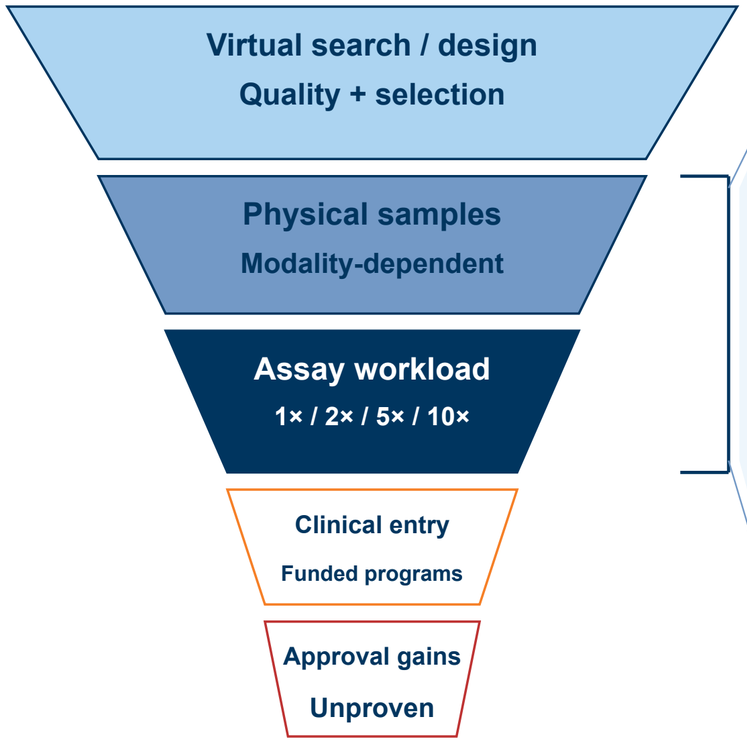
Sources: Lyu 2019; Sumitomo 2021; BIO 2011–2020; Wong 2000–2015. Workloads are analyst sensitivities, not empirical cohorts.

# AI selection and biological evidence shape tool demand

Workflows vary by modality; paid demand depends on funded assays, depth, net inputs and prices.

## MODALITY-AWARE DISCOVERY WORKFLOW

Schematic widths; workload multipliers are sensitivities




**SMALL MOLECULES**

Fewer syntheses per program are possible.

**PROTEINS / ANTIBODIES**

Pooled screens scale; individual profiling differs.

WET LAB = BUILD + TEST
No universal molecule-count funnel.
AI clinical advantage remains unproven.

## The LST ecosystem behind the capacity gate

Issuer-cited subset estimates · different scopes / years · source conflicts disclosed below

### Molecular construction

DNA / RNA synthesis, enzymes
Chemical building blocks

**RMB 1.33bn**

China • 2025 • DNA synthesis services [1]
Oligos + long fragments only

TWIST BIOSCIENCE · GenScript Make Research Easy

| CHINA / HK LISTINGS | US LISTINGS / PARENTS |
|---|---|
| GenScript · 1548.HK | Twist · TWST |
| Vazyme · 688105.SH | Thermo Fisher · TMO |
| Tsingke · NEEQ:874813 | |

### Protein & functional assays

Recombinant proteins, antibodies
Binding and activity assays

**US$3.5bn**

Global · 2025 · F&S RUO proteins [2]
Issuer-commissioned; estimate disputed

Acro BIOSYSTEMS

| CHINA / HK LISTINGS | US LISTINGS / PARENTS |
|---|---|
| ACRO · 301080.SZ | Bio-Techne · TECH |
| Sino · 301047.SZ | Thermo Fisher · TMO |

### Cells & culture systems

Cell models, media, feeds
Bioreactors and cell expansion

**Source pending**

China • 2024 • bioprocess media [3]
Original filing unavailable; TAM withheld

**Gibco / HyClone**

| CHINA / HK LISTINGS | US LISTINGS / PARENTS |
|---|---|
| OPM · 688293.SH | Gibco / Thermo · TMO |
| ACRO · 301080.SZ | Cytiva / HyClone · DHR |

### Labware & sample handling

Tips, plates, tubes, filtration
Automation-compatible labware

**~RMB 23bn**

China • 2024 • plastic labware [4]
Includes clinical and industrial uses

BIOFIL · BD

| CHINA / HK LISTINGS | US LISTINGS / PARENTS |
|---|---|
| JET Biofil · 688026.SH | Corning · GLW |
| Changhong · 300151.SZ | BD · BDX |

### Measurement & analytics

Sequencing, LC / LC-MS, flow
Imaging and quality control

**RMB 4.3bn**

China • 2024 • sequencing tools [5]
Sequencing subset; 2024 policy baseline

MGI · BD flow → WAT†

| CHINA / HK LISTINGS | US LISTINGS / PARENTS |
|---|---|
| MGI · 688114.SH | Illumina · ILMN |
| NanoMicro · 688690.SH | Agilent · A |
| | Waters · WAT† |

### Purification & scale-up

Analytical columns, process resins
Filtration and single-use systems

**Source pending**

China • 2024 • bioprocess resins [3]
Original filing unavailable; TAM withheld

NanoMicro · cytiva

| CHINA / HK LISTINGS | US LISTINGS / PARENTS |
|---|---|
| NanoMicro · 688690.SH | Cytiva · DHR |
| Sepax · 688758.SH | Repligen · RGEN |

**Automation + data orchestration span every tool family** — TECAN. SIX: TECN

† Ex-BD biosciences / flow moved to Waters on 9 Feb 2026; BD (BDX) here means specimen collection.
Protein TAM is issuer-commissioned / disputed. Media & resin TAMs withheld. Sources, scope and tickers: SOURCES.md.

# Fewer molecules. Deeper experimental data.

AI may reduce candidate counts while increasing characterization per molecule

From molecule throughput to data density: a potential driver of life science tools demand

## Existing data: staged screening

### 1,000 × 1 initial label

Illustrative early single-task dataset

| Antibody | Binds? |
|---|---|
| Ab-001 | Yes |
| Ab-002 | No |
| Ab-003 | Yes |
| Ab-004 | Yes |
| Ab-005 | No |
| Ab-006 | Yes |

**Binding first; deeper tests for selected hits**

Multi-assay characterization already exists

## AI: richer, linked training data

### 300–1,000 antibodies · Up to 9 label types

Measure selected properties on shared molecules; fill the information gaps

| Antibody | KD | kon | koff | expr | Tm | agg | poly | epi | cell |
|---|---|---|---|---|---|---|---|---|---|
| Ab-001 | 120 | 7200 | 1800 | 8400 | 5100 | 75 | 3200 | 9000 | 430 |
| Ab-002 | 8100 | 640 | 5400 | 2200 | 9500 | 1600 | 4100 | 500 | 6900 |
| Ab-003 | 2600 | 9200 | 410 | 6500 | 3100 | 7900 | 950 | 4800 | 8300 |
| Ab-004 | 9800 | 3100 | 7400 | 1200 | 6800 | 380 | 5500 | 1900 | 9100 |
| Ab-005 | 540 | 4700 | 8800 | 3600 | 2000 | 6200 | 10000 | 710 | 5700 |
| Ab-006 | 6300 | 1400 | 3000 | 9900 | 4600 | 8700 | 180 | 7600 | 2400 |

**Learn binding, expression, stability and functional relationships**

Synthetic values: 1–10,000. Darker = larger; no biological units or quality ranking.

## Reuse existing data

Normalize formats, conditions and access
Better training may need no new experiment

## Stage 0: fill data gaps

Add missing assays and biological diversity
Create reusable model-training datasets

## Active learning: measure selectively

Prioritize informative molecules and tests
Limit coverage of costly measurements

## The nine labels: what each property measures

| | | | | | |
|---|---|---|---|---|---|
| **KD** | Equilibrium binding strength | **kon** | How quickly binding forms | **koff** | How quickly the complex separates |
| **expression** | Antibody yield from a cell system | **Tm** | Thermal unfolding transition | **aggregation** | Molecules forming larger clusters |
| **polyreactivity** | Binding to multiple non-targets | **epitope** | The binding site on the target | **cell potency** | Functional activity in living cells |

Illustrative counts. Nine label types are not nine independent assays; richer data need not come at unchanged cost.

Sources: Jain et al., 2017; Arsiwala et al., 2025; van Niekerk et al., 2026. Mechanism illustration; definitions and links in SOURCES.md.

# Experimental data density and the LST ecosystem

Stage 0: biological data production creates a wet-lab use case

Learning developability requires comparable assay data and an integrated tool workflow

## Stage 0: biological data production

Continuous input to training and iteration

Binding kinetics
**KD / kon / koff**

Expression & physicochemical properties
**expression / Tm / aggregation**

Specificity & cell function
**polyreactivity / epitope / cell potency**

Training records: sequence, target, conditions, raw curves, replicates, batches and failures

**Train / improve** → **Design / validate**

Assays inform models

## Labels → assays → purchases

Tool revenue needs a bridge from label coverage to workflow volume and unit consumption

### Construction & expression

DNA synthesis, vectors, expression, purification

Expression labels + samples for other assays

Count shared sample preparation once

### Protein & functional testing

SPR / BLI, binding curves, epitope binning

Target proteins, sensors / chips, buffers

KD / kon / koff can share one kinetic run

### Cells & culture systems

Functional cells, media, transfection, expansion

Cell potency; selected polyreactivity assays

Record cell model, target level and conditions

### Measurement & analysis

SEC / HIC, nanoDSF, LC-MS, flow cytometry

Stability, aggregation, characterization / QC

Columns are reused; model instrument CAPEX separately

### Sample handling, labware & automation

Plates, tips, tubes, filtration / beads, liquid handling

Across all workflows; miniaturization, reuse and scheduling alter net consumption

**ACROBiosystems: target proteins, functional cells & characterization**

AIDD-Ready offering exists; paid orders and repeat purchases still need verification

**NanoMicro: analytical separation & lab purification materials**

Track sample runs, column life and ASP; manufacturing resin demand has a longer lag

**LST purchases ≈ Σ [workflow executions × net purchased inputs per run × ASP]**

Instrument CAPEX is separate. Product alignment does not establish AI orders or revenue. Sources: Ginkgo, Sartorius and company websites.

# Typical protein / antibody screening workflow

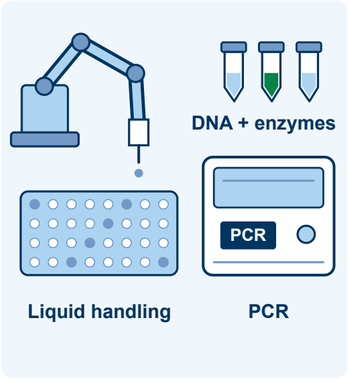
Flat experimental schematics: the tools, inputs and suppliers behind each step

## BUILD · Make physical samples

### 01 / BUILD

## Construct design / DNA synthesis

Make DNA constructs for the selected candidates




**Sequence-verified DNA**

**REPRESENTATIVE SUPPLIERS**

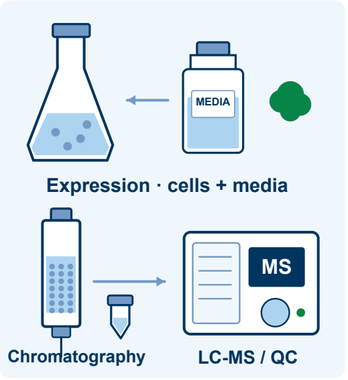
DNA + enzymes
Twist · GenScript · Vazyme

Liquid handling / PCR
Tecan · Hamilton · Thermo Fisher

### 02 / BUILD

## Protein expression & purification

Express the protein, purify it and check sample quality




**Purified protein + yield**

**REPRESENTATIVE SUPPLIERS**

Expression / media
GenScript · Gibco

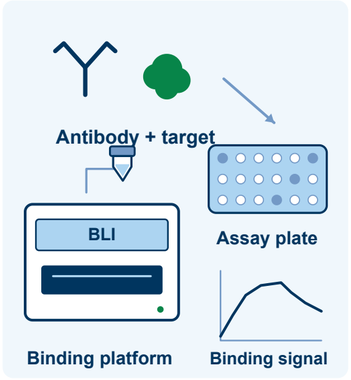
Separation / QC
Cytiva · NanoMicro · Waters

## TEST · Generate biological evidence

### 03 / TEST

## Primary binding screen

Identify binders against the chosen target protein




**Binding hits**

**REPRESENTATIVE SUPPLIERS**

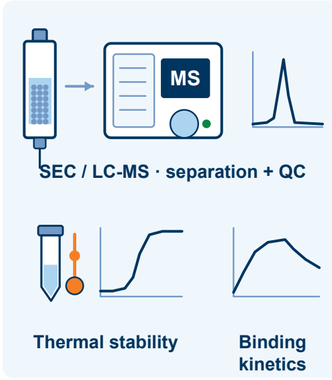
Target reagents
ACROBiosystems · Sino Biological

Binding platforms / labware
Sartorius · Cytiva · JET Biofil

### 04 / TEST

## Multi-property characterization

Measure binding, stability and developability risks




**Linked multi-property labels**

**REPRESENTATIVE SUPPLIERS**

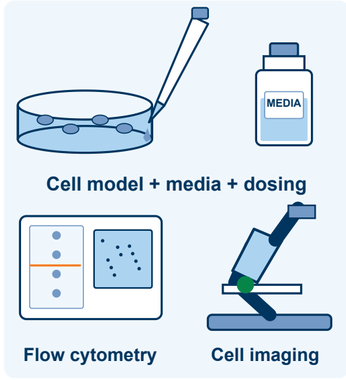
Separation / analytics
Waters · Agilent · NanoMicro

Stability / binding
NanoTemper · Sartorius

### 05 / TEST

## Cell-function validation

Test biological activity in a relevant cell model




**Cell potency**

**REPRESENTATIVE SUPPLIERS**

Cell inputs / labware
Gibco · Bio-Techne · Corning

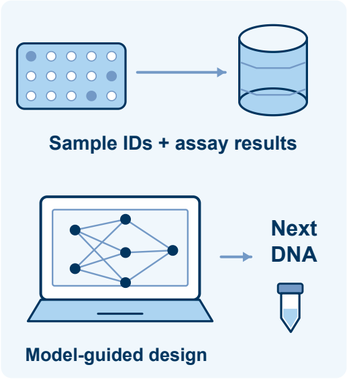
Flow / imaging
Waters Biosciences* · Mol. Devices

## LEARN · Iterate

### 06 / LEARN

## Data integration & feedback

Link samples and results to the next model-guided design




**Traceable training data**

**REPRESENTATIVE SUPPLIERS**

Automation
Tecan · Hamilton

ELN / LIMS / data
Benchling · Genedata

**Linked results → next design → next experiment**

**Automation + data orchestration span every step**

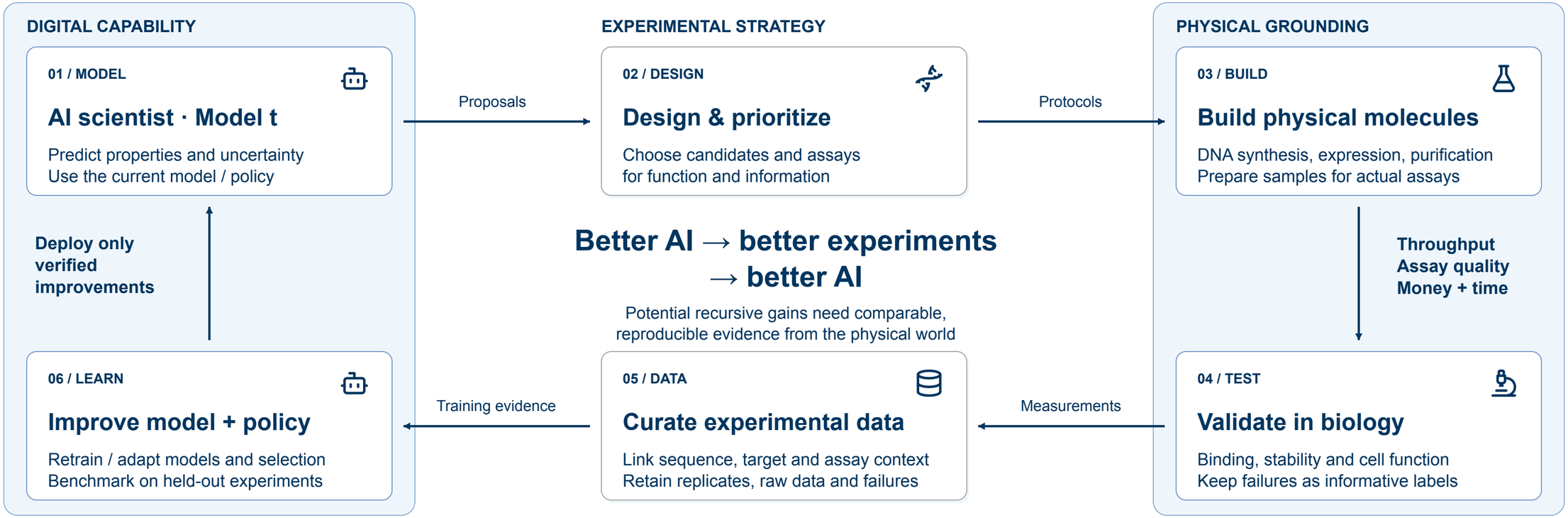
Representative experiments, not a fixed protocol. Assays may run in parallel. LC-MS is shown for analytical QC.

*Waters Biosciences is the former BD Biosciences portfolio. Flat schematics and text are editable; supplier examples and sources: workflow/SOURCES.md.

# RSI: AI design and biological validation

How experimental feedback could improve both biological designs and the AI design process

**RECURSIVE SELF-IMPROVEMENT · CONCEPTUAL MECHANISM**




## LST supports every return to physical biology

**BUILD**
DNA synthesis · expression · purification

**TEST**
Proteins · biosensors · analytical columns
Cell assays · measurement platforms

**CONNECT + SCALE**
Labware · automation · data platforms
Comparable records across batches

**The pace of improvement depends on useful experimental evidence per cycle**

## Potential LST demand ∝ funded cycles × assay depth × net inputs per assay

Automation and miniaturization can reduce inputs per measurement; instrument CAPEX is separate.

Conceptual RSI mechanism. Closed-loop learning does not establish sustained autonomous RSI. Primary sources: SOURCES.md.